

PROJECT PROPOSAL

Food Delivery Performance & Analytics

Data Analysis & Visualization

Objective

To analyze real-world food delivery data in order to uncover meaningful patterns in delivery times, customer ratings, and operational performance. The project aims to identify key factors that drive delays, low satisfaction scores, and inefficiencies — and to present these findings through an interactive dashboard that could support data-driven decisions for delivery platforms and restaurant partners.

Project Description

Food delivery platforms generate enormous volumes of data daily — order timestamps, delivery durations, driver routes, customer feedback, and restaurant metrics. This project focuses on collecting, cleaning, and analyzing such datasets to surface actionable insights. Using publicly available datasets (e.g., from Kaggle or simulated platform exports), the team will build a complete data pipeline from raw ingestion to final interactive dashboard.

Key research questions include:

- What are the average and distribution of delivery times across different times of day, days of week, and weather conditions?
 - Which factors most strongly correlate with low customer ratings — delay, order accuracy, or restaurant type?
 - Are there geographic or temporal patterns in delivery performance?
 - How do different restaurants or cuisine types compare in fulfillment speed and satisfaction?
 - What operational bottlenecks can be identified from the data, and what improvements can be recommended?
-

Tools & Design Approach

Data Collection

- Kaggle food delivery datasets (Zomato)
- Simulated or anonymized platform export data in CSV format
- Web scraping (if needed) for supplementary restaurant metadata

Visualization & Reporting

- Power BI — interactive dashboard with KPIs, trend charts, geographic views, and filters
- Final written report documenting methodology, insights, and recommendations

Technologies

Category	Tools
BI & Dashboards	Microsoft Power BI
Spreadsheets	Microsoft Excel
Data Sources	Kaggle Datasets, CSV exports, web scraping
Version Control	Git / GitHub

Team Members & Responsibilities

Name	Role & Responsibilities
Youssef Hussein Sophy	Data Collection & Report Writing — sourcing datasets, API/scraping, final documentation
Mohanad Khaled Elsayed	Data Cleaning — missing values, deduplication, formatting standardization
Mohamed Ahmed Abu Alyazeed	Data Modeling — feature engineering, statistical analysis, aggregation logic
Mohamed Hassan Nour	Data Visualization — EDA charts, Power BI visual design, insight storytelling
Omar Khalifa Kheder	Dashboard Organization — layout, UX of the final dashboard, presentation prep

Project Timeline — 4-Week Plan

Week 1: Data Collection & Understanding

Team Member	Task
Youssef	Source and download food delivery datasets from Kaggle and open data portals
Youssef	Explore dataset fields: order time, delivery time, ratings, distance, weather, etc.
Mohamed A.	Define the data schema and analysis-ready structure
All Team	Kickoff meeting: align on scope, set up GitHub repo, assign week tasks
Mohanad	Initial data inspection — identify missing values, encoding issues, outliers
Deliverables	Raw dataset(s) collected from at least 2 sources
	GitHub repository initialized with project structure
	Data schema and field definitions document
	Initial inspection notes (data quality observations)

Week 2: Data Cleaning & Preparation

Team Member	Task
Mohanad	Handle missing values: impute or flag delivery time and rating fields
Mohanad	Remove duplicates; resolve inconsistencies in restaurant and location names
Mohamed A.	Standardize date/time formats; engineer new features (e.g., time-of-day, delay flag)
Youssef	Cross-check data against secondary sources where possible
Mohamed A.	Export cleaned dataset
Deliverables	Cleaned and merged dataset (CSV format)
	Data quality report (records removed, imputed, transformed)

Week 3: Exploratory Data Analysis & Visualization Design

Team Member	Task
Mohamed H.	Perform EDA: delivery time distributions, rating trends, peak hours analysis
Mohamed H.	Identify top correlations: what most affects delivery speed and ratings?
Omar	Design Power BI dashboard layout — define KPIs, pages, and filter structure

Omar	Build draft visuals: delivery time histogram, rating heatmap, performance by restaurant
Youssef	Begin writing the analysis section of the final report
Deliverables	Draft Power BI dashboard (at least 4 interactive visuals)
	Summary of key findings to highlight in the final report

Week 4: Dashboard Finalization & Reporting

Team Member	Task
Omar	Refine all Power BI visuals; add slicers, tooltips, and drill-through filters
Mohamed H.	Polish chart aesthetics and ensure consistent color coding and labels
Youssef	Complete the final project report (methodology, findings, limitations, recommendations)
All Team	Internal review: test dashboard interactivity, proofread report, fix errors
All Team	Prepare and rehearse final presentation
Deliverables	Final interactive Power BI dashboard
	Complete project report (PDF format)
	Presentation deck (PowerPoint or PDF)
	All files uploaded and organized on GitHub

Expected Outcomes

- A clean, analysis-ready food delivery dataset with engineered features for performance metrics
- An interactive Power BI dashboard showing delivery time trends, rating distributions, and restaurant comparisons
- Identification of the key drivers of delivery delays and low customer satisfaction
- Data-backed recommendations for improving delivery performance and customer experience
- A complete project report and presentation suitable for academic submission